

Test Case Log

Team Predictrix

Ashen Era Archive Assistant — Sub-track 1C — SLIIT Codefest 2026

Every command below can be copied and run directly against the submitted repository. All results shown were captured from real terminal runs during development, not invented for this document.

One-time setup

```
python -m src.ingest.parse_corpus
```

```
python -m src.embed.embed_and_store
```

1. Official Sub-track 1C sample questions

Our submitted track. These two questions are the actual official samples.

1C-1 — Gloamreach founding year

```
python -m src.agent.agent_loop "State the precise year in the Age of Shadows that marks the true founding of Gloamreach."
```

Answer	246 AS
Confidence	High
Iterations	3
Conflict	Codex (0.9) claims explicit authority for 246 AS, correcting popular accounts. Ephemera (0.3) claims 286 AS. Wiki (0.7) notes it is contested. Codex's explicit claim wins.

1C-2 — Gauntlet of Sorrowfell forging year

```
python -m src.agent.agent_loop "In which year was the 'Gauntlet of Sorrowfell' actually forged?"
```

Answer	391 AS
Confidence	High
Iterations	2
Conflict	Codex overrides ephemera's competing claim of 360 AS.

2. Official Sub-track 1B sample questions (generalization check)

Not our submitted track. Run to test whether the 1C search loop generalizes to genuine multi-document questions without dedicated 1B engineering.

```
python -m src.agent.batch_test --track 1B
```

ID	Question	Answer	Iter.
1b_007	Accord won by Ederon Fellgard's faction	The Leaden Accord	4
1b_006	Member of the faction that won the War of Drowned Light	Ignatz Fellgard	4

1b_022	War won by Ravena Stormwell's faction	The War of Drowned Light	4
1b_013	Dominion encompassing the Gravemaw Wyrms' lair	The Bleeding Crown	4
1b_005	War won by Isolde Mournvale's organization	The War of Drowned Light	4
1b_009	Halvard Crowhurst's connection to Purge of Blackport victors	Via membership in The Iron-Ring Cartel	4
1b_003	Redoubt holding Cerys Sablewood's relic	Gloamreach	5

Result: 7/7 answered, 0 errors. Every question took 4–5 search iterations, confirming genuine multi-step chaining rather than a single lucky lookup.

Why a short final answer still proves multi-hop reasoning: the reasoning trace for 1b_007 shows the actual path taken — no single document states the answer directly:

```
Layer 0: search "Ederon Fellgard" -> found: member of The Iron-Ring Cartel Layer 1: search "Iron-Ring Cartel accord won" -> found: won The Leaden Accord Layer 2: answer -> "The Leaden Accord"
```

3. Custom stress-test questions

3-1 — Honest uncertainty check

```
python -m src.agent.agent_loop "What is the exact hex color of Ashreach's official flag?"
```

Answer	Insufficient evidence to determine this.
Confidence	Low
Iterations	5 (max reached)
Purpose	This fact does not exist in the corpus. Tests whether the agent admits it doesn't know instead of fabricating an answer. Passed.

3-2 — Garbage input robustness check

```
python -m src.agent.agent_loop "test"
```

Answer	Insufficient evidence to determine this.
Purpose	Confirms no crash or hallucination on a non-question input. Passed.

4. Live demo budget: how many questions can we safely show?

OpenRouter's free tier allows 50 API requests per day. Each search iteration costs 2 calls (one to embed the query, one for the model's decision), plus 1 extra call for the conflict-resolution check whenever a question's answer draws on 2 or more distinct sources, which was true for every question we tested.

The cost formula

```
cost(question) = (iterations x 2) + 1 [if 2+ sources are used, true in all our tests]
```

This formula is not a guess. It matches real behavior exactly: our 7-question 1B batch used 29 total iterations, and the formula predicts $29 \times 2 + 7 = 65$ calls. The very next test that day failed immediately with a rate-limit error, which is only possible if fewer than 65 of the day's 50 calls remained, confirming the formula's accuracy.

Cost of each candidate demo question, using real recorded iteration counts

Question	Iterations	Cost (calls)
1C-1: Gloamreach founding	3	$3 \times 2 + 1 = 7$
1C-2: Gauntlet of Sorrowfell	2	$2 \times 2 + 1 = 5$
1B example: Ederon / Leaden Accord	4	$4 \times 2 + 1 = 9$
Honest uncertainty: Ashreach flag	5 (max)	$5 \times 2 + 1 = 11$

The recommended demo set

Running all four questions above (the two official 1C questions, one 1B generalization example, and one honest-uncertainty example) costs $7 + 5 + 9 + 11 = 32$ calls, leaving **18 calls** of the 50-call daily allowance as a safety buffer for retakes if a search needs to be re-run during recording.

This set of four is deliberately chosen to demonstrate every core capability once each: conflict resolution (both 1C questions), multi-hop chaining across documents (the 1B example), and honest refusal to guess (the uncertainty example). Running more than this does not add new evidence of capability, only repeats what has already been shown, so four is both the safe and the sufficient number for a live recording.

If more room is needed

Two options exist if additional questions or retakes are required beyond the 18-call buffer: wait for the daily reset and split recording across two sessions, or add a one-time \$10 OpenRouter credit top-up, which raises the daily limit to 1000 requests and removes this constraint entirely for the remainder of the project.